Multi-Disease Detection Report

CNN + LLM contextual reasoning • generated 2026-08-24 23:02

Patient context

Age	72
Sex	Male
Symptoms	Short-term memory loss, repeatedly asking the same questions, forgetting recent conversations, difficulty finding words
History	72-year-old woman, gradual memory decline over 2 years, increasingly relies on family members for reminders

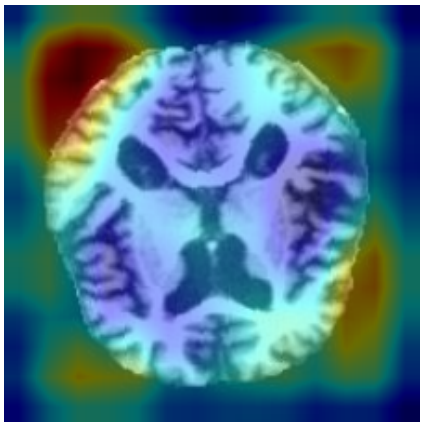
Model prediction

Alzheimers_ModerateDemented Neurodegenerative

Alzheimer's Disease (dementia staging) · Neurological / Brain · Brain MRI

#	Condition	Confidence
1	Alzheimers_ModerateDemented	96.9%
2	kidney_normal	0.9%
3	Alzheimers_MildDemented	0.6%
4	Alzheimers_VeryMildDemented	0.4%
5	Alzheimers_NonDemented	0.2%

Grad-CAM (model attention)



LLM contextual interpretation

Model suggests Alzheimer's Disease (dementia staging) (Alzheimers_ModerateDemented) at 96.9% confidence.

Model finding. The CNN's top class is 'Alzheimers_ModerateDemented' (Neurodegenerative, Neurological / Brain) on Brain MRI, at 96.9%.

Confidence. High model confidence (96.9%) with a clear 96.1 pt margin over the runner-up.

Differential

- **Alzheimers_ModerateDemented** — ranked #1 at 96.9%
- **kidney_normal** — ranked #2 at 0.9%
- **Alzheimers_MildDemented** — ranked #3 at 0.6%

Grad-CAM correlation. Model attention was a moderately sized region in the upper-center portion of the image (~19% high-salience area). Verify this region is anatomically plausible for Brain MRI.

Biomedical context. [KB] More pronounced cortical/hippocampal atrophy consistent with moderate dementia. Group: Alzheimer's Disease (dementia staging); organ: Neurological / Brain; modality: Brain MRI; category: Neurodegenerative (severity: high). Alzheimer's disease shows progressive medial-temporal and hippocampal atrophy on MRI, correlating with clinical dementia staging from non-demented to moderate. [BioGPT fallback — set ENABLE_BIOGPT=true to run the real model] Clinical assessment plus structural MRI supports staging; biomarkers (CSF, amyloid PET) and cognitive testing aid confirmation. [Med-PaLM 2: PLACEHOLDER — no public API; see medpalm_adapter.py to wire MedLM]

Patient considerations. Provided context (age: 72; sex: Male; symptoms: Short-term memory loss, repeatedly asking the same questions, forgetting recent conversations, difficulty finding words; history: 72-year-old woman, gradual memory decline over 2 years, increasingly relies on family members for reminders) should be correlated with the finding.

Recommended next steps

- Correlate with the full clinical picture and prior imaging.
- Obtain gold-standard confirmation appropriate to the modality (e.g., biopsy/histology or specialist review).
- Escalate to the relevant specialist if the category is malignant or high-severity.

Uncertainty & limitations

- Interpretation generated WITHOUT the Claude reasoning model (offline fallback).
- Model trained on specific public datasets; performance may not transfer to other scanners/populations.
- A confident softmax score is not calibrated probability of disease.

■ Research decision-support only — not a medical diagnosis. All findings require confirmation by a licensed clinician.